

10 Capstone projects

- 1. Autonomous Research Assistant** An AI agent that browses the web, reads papers, synthesizes findings, and generates a structured research report — with citations — on any topic, using tool-calling and RAG pipelines.
 - 2. Multi-Agent Software Engineering System** A team of specialized agents (planner, coder, reviewer, tester) that collaboratively build, debug, and deploy a working application from a natural language specification.
 - 3. AI-Powered Personal Finance Manager** An agent that connects to bank APIs, categorizes transactions, detects anomalies, forecasts budgets, and proactively alerts users with actionable recommendations.
 - 4. Autonomous Clinical Decision Support Agent** A deep learning system trained on medical records and imaging data that assists doctors by suggesting diagnoses, flagging risks, and summarizing patient history — with explainability layers.
 - 5. Self-Improving Code Generation Agent** An agent that writes code, runs it in a sandbox, observes errors, reflects on failures, and iteratively rewrites until tests pass — implementing a full ReAct / reflection loop.
 - 6. Agentic E-Commerce Operations Bot** An end-to-end agent managing an online store: monitoring inventory, auto-negotiating with suppliers via email, adjusting pricing based on competitor scraping, and handling customer queries.
 - 7. Deep RL Agent for Adaptive Game Playing** A reinforcement learning agent trained via self-play and curriculum learning to master a complex game (e.g., a custom RTS or puzzle environment) and adapt its strategy to new opponents.
 - 8. Multimodal Document Intelligence Agent** An agent that ingests PDFs, images, spreadsheets, and contracts — extracts structured data using vision + language models — and answers complex cross-document questions autonomously.
 - 9. AI Safety & Red-Teaming Agent** An agentic system that automatically probes other LLMs for jailbreaks, prompt injections, and bias — logs vulnerabilities, classifies severity, and generates mitigation reports.
 - 10. Autonomous Urban Planning Simulator** A deep learning + agent-based simulation where AI agents representing citizens, planners, and businesses interact in a city model — optimizing for traffic, emissions, and livability using reinforcement learning.
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Each of these combines **agentic orchestration** (planning, tool use, memory, reflection) with **deep learning** components (transformers, RL, vision models), making them strong capstone candidates for graduate or advanced undergraduate programs. Want me to flesh out any of these with architecture diagrams, tech stacks, or implementation roadmaps?